

LE PUY LOUDES AP, France

WMO: 074710

Lat: 45.0745N

Lon: 3.7641E

Elev: 2731

StdP: 13.30

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.6	17.8	8.4	9.3	16.7	12.7	11.6	20.3	23.9	40.6	21.2	39.8	4.1	240	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	85.3	63.4	81.8	62.9	78.5	62.1	66.3	79.2	64.8	77.2	63.4	75.2	7.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.9	91.7	71.3	60.3	86.5	69.7	58.9	82.1	68.1	32.5	79.1	31.3	77.1	30.2	75.2	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.7	18.0	15.8	7.1	90.6	5.6	3.6	3.1	93.2	-0.1	95.3	-3.3	97.4	-7.3	100.0	
			6.3	69.0	5.4	1.3	2.5	69.9	-0.7	70.7	-3.7	71.4	-7.6	72.3	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	34.1	35.1	41.0	45.5	52.3	59.8	63.7	62.8	56.0	50.1	40.7	35.7
	DBStd	12.36	7.29	7.81	6.39	6.00	6.02	6.41	5.59	5.79	5.65	6.76	7.24	7.54
	HDD50	2229	494	418	284	159	45	3	0	0	12	82	287	445
	HDD65	6277	959	838	745	584	393	180	90	113	275	461	730	909
	CDD50	1551	0	0	3	25	117	297	426	398	191	86	7	1
	CDD65	124	0	0	0	0	1	24	50	46	3	0	0	0
	CDH74	1924	0	0	0	1	43	384	719	676	96	5	0	0
	CDH80	509	0	0	0	0	3	94	197	207	8	0	0	0
Wind	WSAvg	6.1	6.1	6.7	7.1	7.1	6.6	5.8	5.9	5.3	5.1	5.4	6.1	5.9
Precipitation	PrecAvg	32.5	2.0	1.7	1.7	2.8	3.5	3.0	3.0	3.0	3.2	3.4	3.2	2.0
	PrecMax	41.8	4.0	3.7	4.3	5.2	6.3	8.3	7.0	7.5	7.4	5.8	7.8	6.9
	PrecMin	23.7	0.2	0.5	0.4	0.3	1.2	0.8	0.9	1.3	0.8	0.7	1.4	0.5
	PrecStd	4.8	0.9	0.7	0.9	1.3	1.4	1.5	1.3	1.3	1.6	1.4	1.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	60.1	66.2	72.0	78.9	87.6	89.8	90.8	81.0	73.7	62.1	54.7
		MCWB	46.8	46.1	49.9	55.4	59.5	62.9	62.9	62.5	61.7	58.2	53.2	47.7
	2%	DB	50.5	53.8	61.2	66.5	73.9	82.5	85.0	85.2	76.8	68.0	57.3	51.4
		MCWB	45.3	44.6	48.0	52.6	58.6	62.9	63.4	63.4	60.1	57.1	51.4	46.7
	5%	DB	47.6	49.5	56.9	62.4	69.7	78.4	81.4	81.4	72.9	64.6	54.7	49.2
		MCWB	43.4	43.4	46.6	50.6	56.7	62.1	62.9	63.0	59.0	56.2	50.2	45.1
	10%	DB	45.0	46.2	52.9	58.1	65.5	74.5	77.7	77.2	68.7	61.3	52.1	46.9
		MCWB	41.6	41.5	45.0	48.7	54.5	60.8	62.4	62.2	57.9	55.0	48.3	43.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	48.3	52.2	56.7	62.4	67.8	68.1	68.1	65.0	60.9	54.5	49.4
		MCDB	53.1	55.9	61.5	69.0	74.5	81.5	79.7	81.3	74.9	68.4	59.8	53.1
	2%	WB	45.9	45.7	49.4	54.0	59.6	65.5	66.1	65.9	62.3	58.7	52.3	47.3
		MCDB	49.6	51.8	58.1	64.5	71.2	78.1	78.9	78.9	72.5	65.4	56.3	50.7
	5%	WB	43.7	43.8	47.5	51.6	57.6	63.5	64.6	64.4	60.6	57.1	50.4	45.6
		MCDB	47.1	48.8	55.4	60.4	67.5	74.7	77.4	76.8	69.9	63.1	54.2	48.7
	10%	WB	41.7	41.8	45.6	49.3	55.6	61.6	63.1	62.9	58.7	55.3	48.5	43.6
		MCDB	44.8	46.2	52.3	56.7	63.8	72.4	75.6	74.4	66.9	60.8	51.9	46.6
Mean Daily Temperature Range	5% DB	MDBR	13.5	15.3	18.8	19.3	19.9	21.8	23.0	23.2	21.0	17.6	14.2	13.5
		MCDBR	18.4	23.1	27.8	28.4	28.7	29.7	30.5	31.6	30.0	23.5	19.4	18.1
		MCWBR	13.2	15.0	16.6	16.3	15.5	13.9	12.8	13.3	15.2	13.8	13.7	13.3
	5% WB	MCDBR	15.6	20.6	23.9	25.0	25.5	26.3	26.8	27.2	24.8	19.8	17.6	15.5
		MCWBR	11.8	13.9	14.9	14.8	14.6	13.4	12.4	12.9	13.3	12.2	12.8	12.0
Clear-Sky Solar Irradiance	taub	0.279	0.299	0.331	0.364	0.371	0.374	0.369	0.365	0.343	0.333	0.301	0.278	
	taud	2.509	2.462	2.407	2.322	2.347	2.366	2.376	2.396	2.442	2.474	2.516	2.520	
	Ebn at Noon	270	283	286	284	285	283	283	281	279	266	258	259	
	Edh at Noon	21	26	32	38	38	38	37	35	31	26	21	19	
All-Sky Solar Radiation	RadAvg	438	717	1094	1426	1686	1992	1976	1719	1324	833	485	382	
	RadStd	53	91	128	177	163	156	156	121	95	84	60	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page